Best Selling Solar Kits & Industry Types Architecture Specification

End-to-End Operational Lifecycle: Location Setup > Project Types > Best Seller Matrix > Website Config > Storefront

EXECUTIVE SUMMARY & BUSINESS GOAL

Boss ke nirdeshon ke anusar, SolarKits platform par do pramukh naye upgrades implement kiye gaye hain:

1. DYNAMIC INDUSTRY TYPES INTEGRATION: Website ke Category Ribbons me ab dummy "On-Grid/Off-Grid" cards ke sthan par Product Configurations > Project Types se live Industry Types (Solar PV, Agro, EV, BESS, Lighting, Thermal) fetch hote hain. Storefront par EPC kisi bhi Industry par click karke uske andar ke sub-types aur unke combo kits explore kar sakte hain.
2. REAL NATIONAL BEST SELLERS SEEDING: Website Configuration ke Showcase Products me dummy cards ko hatakar live All-India National Best Sellers (3kW DCR Complete Home Kit, 2kW Rooftop Kit, 1kW DCR Kit) automatically feed kar diye gaye hain.
3. ORGANIC DATABASE RANKING: Kits ki ranking real bookings (Units Booked > Orders Count > Revenue) ke anusaar Country, State aur District levels par autonomously update hoti hai.

1. Interconnected Modules Architecture

Module / Tab Name	Admin Portal Path	Key Responsibility & Features	Source Collections
1. Location Setup	Settings > Location Setting > Setup Location	Serviceable States, Districts & 6-digit Pincodes registry.	geolocation_level_1, level_2
2. Project Types Logic	Product Config > Project Types	Master Industry Types, Categories, Subcategories, System Types & Ranges.	sys_industry_types, sys_filter_categories
3. Best Seller Matrix	Solar Kit Config > Best Seller Configuration	Dynamic organic ranking (Units Booked > Orders > Revenue) & manual override (#1).	pc_combo_kits, best_seller_kits, orders
4. Website Showcase	Website Config > Solar Store Website	Live sync of Industry Types cards & 1-click sync of National Best Sellers with 8 field toggles.	website_configurations (solar-store)
5. Solar Storefront	Solar Store (Customer / EPC View)	Dual-mode PIN/District finder + Interactive Industry & Kit Explorer modal.	Direct Public REST APIs

2. End-to-End System Data Flow

DATA LIFECYCLE & SYNC MECHANISM

• Master Creation: Admin "Project Types" me nayi industry (jaise Solar EV ya Solar Agriculture) add karta hai.
• Automatic Ribbon Sync: "Website Configurations > Solar Store" tab me "Live Sync from Project Types" button dabate hi sabhi active industries apne subcategories preview ke sath homepage category ribbons me aa jaati hain.
• National Best Sellers Feed: Database me sabse jyada bikne wali kits (3kW, 2kW, 1kW) website showcase products me pre-fill ho jaati hain.
• Storefront EPC Discovery: Jab EPC website par aata hai, toh Industry Card par click karte hi ek interactive modal khulta hai jo us industry ke andar ke sabhi project categories, system types aur unki live combo kits ko showcase karta hai.

Project Types Hierarchy & Database Schemas

Detailed architectural mapping from Industry Types to Combo Kits

3. The 5-Level Project Types Hierarchy

Hierarchy Level	Schema Collection	Example Values	Storefront Role
Level 1: Industry Type	sys_industry_types	Solar PV, Solar Agriculture, Solar EV, Energy Storage, Solar Lighting, Solar Thermal	Homepage Category Ribbon Cards
Level 2: Project Category	sys_filter_categories	Residential Solar, Commercial & Industrial, Solar Water Pumping, EV Infrastructure	Industry Explorer Category Tabs
Level 3: Sub-Category	sys_filter_subcategories	Individual Home, Housing Society, Submersible Pump, Solar Carport, Factory	Application Use-Case Pills
Level 4: System Type	sys_filter_type_maps	On-Grid, Off-Grid, Hybrid, Zero-Export, DC Submersible, Type-2 AC	Technical Operating Technology
Level 5: Capacity Range	pc_project_ranges	1 - 10 kW, 11 - 50 kW, 3 HP - 10 HP, 5 kWh - 20 kWh	Sizing / Wattage Specifications

4. Master Industry Types Configured in Platform

Industry Type	Slug & Icon	Subcategories / Applications Preview	Typical Pre-Configured Kits
Solar PV Ecosystem	solar-pv (Sun)	Residential Solar • Commercial & Industrial • Ground-Mount	1kW, 2kW, 3kW, 5kW DCR On-Grid & Hybrid Kits
Solar Agriculture	solar-agriculture (Agri)	Solar Water Pumps • Cold Storages • Agro-processing Mills	3HP, 5HP, 7.5HP DC/AC Solar Pump Kits (PM KUSUM)
Solar EV Infrastructure	solar-ev (Zap)	Solar Carports • 2W/3W/4W EV Charging Stations	7.4kW AC Type-2 Charger & 15kW Carport Combos
Energy Storage (BESS)	energy-storage (Box)	Residential BESS • Industrial Peak Shaving • Remote Microgrids	5.12kWh & 10kWh LFP Battery Storage Packs
Solar Lighting & Security	solar-lighting (Grid)	Smart All-in-One Streetlights • Factory Perimeter • Rural Highmasts	40W All-In-One & 90W Semi-Integrated Streetlights
Solar Thermal & Heating	solar-thermal (Sun)	Domestic Hot Water • Commercial Boilers • Dairy Heating	200 LPD ETC & 1000 LPD Industrial FPC Systems

5. Storefront Interactive Industry Explorer Modal

EPC DRILL-DOWN EXPERIENCE

Storefront par Category Section ke card par click karne par naya `IndustryKitExplorerModal` activate hota hai:
• EPC ko us Industry ke sabhi active Project Categories tabs milte hain (e.g. Residential vs Commercial).
• Har category ke andar sub-categories (Individual Home, Housing Society) aur System Types (On-Grid, Hybrid) show hote hain.
• Saath hi us application ke liye tayar Pre-Engineered Combo Kits (Price, Watt, Rating, Badges) dikhte hain jinhe EPC wahi se direct order ya view kar sakta hai.

Website Configuration Sync & National Best Sellers

Replacing legacy dummy cards with live organic best sellers and 8 field toggles

6. Live Feed: Real National Best Seller Combo Kits

Rank & Badge	Kit Title & Specs	Selling Price vs MRP	Organic Metrics & Stock Status
#1 Top Pick in India (Ø=Ý% Bestseller)	3 KW Solarkits DCR Complete Home Combo Kit	¹79,498.75 MRP: ¹98,000 (19% OFF)	184 Units Booked • 98 Orders +P 4.9 (228 Reviews) • In Stock
#2 in India (&¡ High Demand)	2 K.W. Solarkits DCR Rooftop Combo Kit (2kW Kit • SolarKits Prime • On-Grid)	¹57,548.75 MRP: ¹72,000 (20% OFF)	126 Units Booked • 72 Orders +P 4.9 (204 Reviews) • In Stock
#3 in India (+P Top Rated)	1 K.W. Solarkits DCR Kit (1kW Kit • SolarKits Prime • On-Grid)	¹31,324.25 MRP: ¹42,000 (25% OFF)	88 Units Booked • 54 Orders +P 4.9 (180 Reviews) • In Stock

7. Card Field Visibility Controls (Show / Hide Switches)

Card Field Toggle	Key	Action & Business Use Case	Default State
Watt / Capacity	showWatt	Displays "3 kW Kit" pill or hides it for custom industrial kits	ACTIVE (SHOW)
Brand & Category	showBrand	Displays "SolarKits Prime • On-Grid Solar Kit"	ACTIVE (SHOW)
Selling Price	showSellingPrice	Displays active selling price in INR (e.g. ¹79,498.75)	ACTIVE (SHOW)
MRP & Savings	showMrp	Displays strikethrough MRP & calculated savings banner	ACTIVE (SHOW)
Discount %	showDiscount	Displays high-contrast green discount badge (e.g. 20% OFF)	ACTIVE (SHOW)
Review & Ratings	showRating	Displays verified customer star rating score & count	ACTIVE (SHOW)
Stock & Dispatch	showStock	Displays "In Stock • Fast Delivery" status tag	ACTIVE (SHOW)
Add to Cart CTA	showCta	Displays "Order Now" / "View Kit" action button	ACTIVE (SHOW)

8. Organic Sales Ranking Algorithm & Regional Demand

ORGANIC RANKING VS MANUAL OVERRIDE

• Organic Sales Logic: Kits are sorted strictly by:
Units Booked (Sales Volume) > Orders Count > Total Revenue (INR)
• Regional Skew: National level par 3kW #1 rehti hai (PM Surya Ghar subsidy). South India (Andhra Pradesh/Kerala) me 1kW/2kW residential tops. Western India (Gujarat/Rajasthan) me 3kW aur 5kW Commercial tops.
• Manual Override: Agar Admin "Ø=ÜQ Make #1" click karta hai, toh `is_manual_override = true` set hokar manual priority top spot par baithti hai; "Ø=Ý Reset" dabate hi kit wapas natural organic sales rank par aa jaati hai.

API Reference Table & Storefront Cascading Resolution

Complete endpoint contracts, public routes, and cascading waterfall

9. Storefront Cascading Resolution Waterfall

CASCADING WATERFALL FOR STOREFRONT VISITORS

Jab koi EPC ya retail customer Solar Store par Pincode ya State/District daalta hai:

1. DISTRICT OVERRIDE: Check if Admin tagged a kit for that exact District (e.g. "Ø=Ý% #1 Best Seller in Rajkot").
2. STATE OVERRIDE: If no district tag, check State-level tag (e.g. "&¡ High Demand in Gujarat").
3. ALL INDIA NATIONAL TAG: If no state tag, display National Best Seller ("& #1 Top Pick in India").
4. DYNAMIC ORGANIC FALLBACK: If no manual tags exist, catalog kits rank automatically by live sales volume.

10. Core REST API Endpoints

Method & Endpoint	Module / Handler	Access Scope	Purpose & Output Payload
GET /admin-api/project-types/public-hierarchy	project.types.handler.js	Public (No Auth)	Returns complete 5-level industry hierarchy, categories, and ranges for the
GET /admin-api/website-content/public/solar-store	website.content.handler.js	Public (No Auth)	Returns landing page content with Industry Category ribbons & seeded
PUT /admin-api/website-content/solar-store	website.content.handler.js	Admin (Auth)	Updates customized industry ribbons, featured products cards and field visibility
GET /admin-api/best-seller-configs	best_seller_kits.handler.js	Admin (Auth)	Unified list of ranked kits with units booked, orders count, revenue & rank
POST /admin-api/best-seller-configs	best_seller_kits.handler.js	Admin (Auth)	Creates or updates regional best seller tag with manual override support.
GET /shop/best-sellers	shop.handler.js	Public / EPC	Returns curated or dynamic sales-ranked best seller combo kits based on customer Pincode/District.